ECON 628

Problem Set #2 — Solutions

Fall 2026

Question 1: Weighted Least Squares and OverID

The build do-file leaves 56,182 workers in 48 age × education × sex cells.

(a) Column (1) of the table below.

(b) Collapse to cell means $\bar{y}_c$, cell counts N_c and within-cell variances s_c^2.

(c) Column (2). The coefficients are identical: the regressors are constant within cells, so the micro OLS normal equations only depend on cell sums, which WLS with weights N_c reconstructs. The standard errors are larger because the cell regression estimates σ^2 from the 48 cell means only (38 degrees of freedom), discarding all within-cell variation.

Log hourly wage on main effects

	(1) Micro OLS	(2) WLS on cell means
Age 25–35	0.3145	0.3145
	(0.0075)	(0.0280)
Age 35–45	0.4580	0.4580
	(0.0073)	(0.0274)
Age 45–55	0.4937	0.4937
	(0.0077)	(0.0289)
Age 55–65	0.4718	0.4718
	(0.0097)	(0.0362)
Age >65	0.2948	0.2948
	(0.0165)	(0.0618)
High school	0.2647	0.2647
	(0.0075)	(0.0282)
Some college	0.4150	0.4150
	(0.0076)	(0.0286)
BA+	0.7714	0.7714
	(0.0078)	(0.0293)
Female	-0.2614	-0.2614
	(0.0046)	(0.0172)
Constant	1.8917	1.8917
	(0.0078)	(0.0292)
Observations	56,182	48

(d) If wages are i.i.d. within cell c with variance σ_c^2,

$$\mathrm{Var}\left(\frac{1}{N_c}\sum_{i=1}^{N_c} y_{ic}\right) = \frac{1}{N_c^2}\sum_{i=1}^{N_c}\mathrm{Var}(y_{ic}) = \frac{\sigma_c^2}{N_c}.$$

(e) Cell by cell, without imposing a common variance:

$$\hat{v}_c = \frac{s_c^2}{N_c}, \qquad s_c^2 = \frac{1}{N_c - 1}\sum_{i\in c}(y_{ic} - \bar{y}_c)^2.$$

(f) WLS of $\bar{y}_c$ on the main effects with weights $1/\hat{v}_c$.

(g) The over-id statistic is the weighted sum of squared residuals,

$$Q = \sum_{c=1}^{48} \frac{\left(\bar{y}_c - x_c'\hat{\beta}\right)^2}{\hat{v}_c} \xrightarrow{d} \chi^2_{48-K}$$

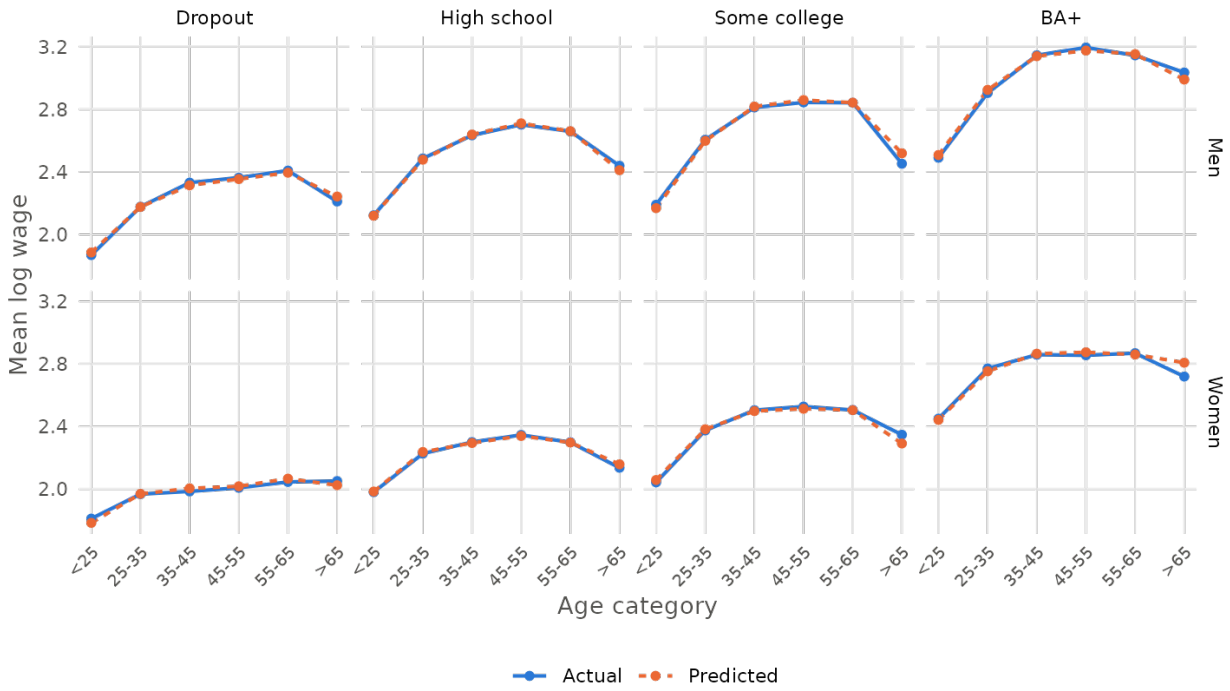
under correct specification. With $K = 10$, $Q = 538.55$ on 38 degrees of freedom, $p < 0.001$. The main-effects model is rejected.

(h) –(i) With all two-way interactions ($K = 33$), $Q = 33.36$ on 15 degrees of freedom, $p = 0.004$. Still rejected at 5%.

Over-identification tests

Specification	Parameters	df	Q	p
Main effects	10	38	538.55	< 0.001
+ two-way interactions	33	15	33.36	0.004
+ two-way, + BA × age × sex	38	10	16.60	0.084

(j)



Actual and predicted cell means, two-way interaction model

The misfit is concentrated among the BA+ group: young college-educated men are over-predicted and older ones under-predicted, with the signs reversed for women. The age profile of the college-educated differs by sex in a way a common age $\times$ sex interaction cannot capture.

(k) Adding a BA $\times$ age $\times$ sex interaction gives $Q = 16.60$ on 10 degrees of freedom, $p = 0.084$: not rejected at 5%.

Question 2: MLE

Write $\Lambda_i \equiv \Lambda(X_i'\beta)$. First note

$$\frac{\partial \Lambda(X_i'\beta)}{\partial \beta} = \frac{e^{X_i'\beta}(1+e^{X_i'\beta}) - e^{X_i'\beta}e^{X_i'\beta}}{(1+e^{X_i'\beta})^2} X_i = \Lambda_i(1-\Lambda_i)X_i.$$

(a)

$$\begin{aligned}\frac{\partial l_i}{\partial \beta} &= \frac{Y_i}{\Lambda_i}\Lambda_i(1-\Lambda_i)X_i - \frac{1-Y_i}{1-\Lambda_i}\Lambda_i(1-\Lambda_i)X_i \\ &= Y_i(1-\Lambda_i)X_i - (1-Y_i)\Lambda_i X_i = \big(Y_i - \Lambda(X_i'\beta)\big)X_i.\end{aligned}$$

(b) β_{ML} solves $\mathbb{E}\big[(Y_i - \Lambda(X_i'\beta))X_i\big] = 0$. By iterated expectations,

$$\mathbb{E}\Big[\big(\mathbb{E}[Y_i \mid X_i] - \Lambda(X_i'\beta)\big)X_i\Big] = 0:$$

the approximation error of $\Lambda(X_i'\beta)$ to the CEF $\mathbb{P}(Y_i = 1 \mid X_i)$ is orthogonal to X_i, as in OLS but with a nonlinear link. This holds at β_{ML} whether or not the logit is correct. If it is correct, $\mathbb{E}[Y_i \mid X_i] = \Lambda(X_i'\beta_0)$ and $\beta_{ML} = \beta_0$; if not, β_{ML} is the best logit approximation to the CEF.

(c) From the first-order condition of $\min_\beta \mathbb{E}\big[(Y_i - \Lambda(X_i'\beta))^2\big]$,

$$\mathbb{E}\Big[\big(Y_i - \Lambda(X_i'\beta)\big)\,\Lambda_i(1-\Lambda_i)\,X_i\Big] = 0.$$

The same prediction error, now orthogonal to X_i weighted by $W_i = \Lambda_i(1-\Lambda_i)$.

(d) When the model is correctly specified, $\mathbb{E}[Y_i \mid X_i] = \Lambda(X_i'\beta_0)$. By iterated expectations both moment conditions become

$$\mathbb{E}\Big[\big(\mathbb{E}[Y_i \mid X_i] - \Lambda(X_i'\beta)\big)\,g(X_i)\Big] = 0,$$

with $g(X_i) = X_i$ for ML and $g(X_i) = W_iX_i$ for NLLS. At β_0 the first factor is zero for every X_i, so both hold and $\beta_{ML} = \beta_{NLLS} = \beta_0$. Under misspecification the two weightings generally deliver different parameters.

(e) The Hessian is

$$\frac{\partial^2 l_i}{\partial\beta\,\partial\beta'} = -\Lambda_i(1-\Lambda_i)X_iX_i',$$

and by the information matrix equality

$$\mathrm{AVar}\big[\sqrt{N}(\hat\beta_{ML} - \beta)\big] = \mathbb{E}\big[\Lambda_i(1-\Lambda_i)X_iX_i'\big]^{-1} \equiv H^{-1}.$$

(f) Without the information matrix equality, the sandwich:

$$\mathrm{AVar}\big[\sqrt{N}(\hat\beta_{ML} - \beta_{ML})\big] = H^{-1}B\,H^{-1}, \qquad B = \mathbb{E}\Big[\big(Y_i - \Lambda(X_i'\beta_{ML})\big)^2 X_iX_i'\Big].$$

Under correct specification $\mathbb{E}[(Y_i - \Lambda_i)^2 \mid X_i] = \Lambda_i(1-\Lambda_i)$, so $B = H$ and this collapses to (e).

Question 3: SML

(a) Conditional on b_i the model is a logit, $\mathbb{P}(Y_i = 1 \mid X_i, b_i) = \mathbb{P}(\varepsilon_i > -b_iX_i) = \Lambda(b_iX_i)$.

Integrating out $b_i = \mu + \sigma\tau$, $\tau \sim N(0,1)$, by simulation with draws τ_{is}:

$$\hat{P}_i = \frac{1}{S}\sum_{s=1}^{S} \Lambda\big((\mu + \sigma\tau_{is})X_i\big), \qquad \hat{\ell} = \sum_{i=1}^{N}\Big[Y_i \ln \hat{P}_i + (1 - Y_i)\ln(1 - \hat{P}_i)\Big].$$

Let $\lambda_{is} = \Lambda((\mu + \sigma\tau_{is})X_i)$. Then

$$\frac{\partial \hat{P}_i}{\partial \mu} = \frac{X_i}{S}\sum_s \lambda_{is}(1 - \lambda_{is}), \qquad \frac{\partial \hat{P}_i}{\partial \sigma} = \frac{X_i}{S}\sum_s \lambda_{is}(1 - \lambda_{is})\tau_{is},$$

and

$$\frac{\partial \hat{\ell}}{\partial \theta} = \sum_i \frac{Y_i - \hat{P}_i}{\hat{P}_i(1 - \hat{P}_i)}\,\frac{\partial \hat{P}_i}{\partial \theta}, \qquad \theta \in \{\mu, \sigma\}.$$

(b) –(c) Draw V once and hold it fixed throughout the optimisation.

(d) BFGS with the analytic gradient gives $\hat{\mu} = 1.551$ and $\widehat{\ln\sigma} = 0.469$ ($\hat{\sigma} = 1.598$), $\hat{\ell} = -1057.08$. Starting values $(0, 0)$, $(0.5, \ln 0.5)$ and $(2, \ln 2)$ all converge to the same point; from $(-1, \ln 3)$ the optimiser drifts to a point with a much lower likelihood, where every λ_{is} saturates at 0 or 1.

Simulated maximum likelihood ($N = 2{,}000$, $S = 1{,}000$)

	Estimate	SE, inverse Hessian	SE, sandwich
μ	1.5507	0.2503	0.2326
$\ln\sigma$	0.4687	0.1975	0.1819

(e) Under correct specification, $\widehat{\mathrm{Var}}(\hat{\theta}) = \hat{H}^{-1}$, with $\hat{H}$ the Hessian of the negative simulated log-likelihood at the optimum.

(f) Sandwich:

$$\widehat{\mathrm{Var}}(\hat{\theta}) = \hat{H}^{-1}\Big(\textstyle\sum_i \hat{s}_i\hat{s}_i'\Big)\hat{H}^{-1}, \qquad \hat{s}_i = \frac{\partial \hat{\ell}_i}{\partial \theta}\Big|_{\hat{\theta}}.$$

The two sets of standard errors are within 8% of each other.

Question 4: Cluster Robust Standard Errors

Standard errors and rejection rates

Parts (a)–(b): standard error of $\hat{\beta}$			
	Homoskedastic	Robust	Clustered
$\sigma_\eta^2 = 1$, $\hat{\beta} = 0.850$	0.0316	0.0316	0.1040
$\sigma_\eta^2 = 0$, $\hat{\beta} = 0.999$	0.0284	0.0284	0.0298
Parts (d)–(e): rejection rate, 1,000 replications, 5% level			
	Robust	Clustered	Block bootstrap
(d) η_c redrawn, $H_0 : \beta = 1$	54.0%	4.7%	5.0%
(e) η_c fixed, $H_0 : \beta = \bar{\eta}$	4.6%	0.0%	0.0%

(a) $\hat{\beta} = 0.850$. Homoskedastic SE 0.0316, robust SE 0.0316, clustered SE 0.1040.

(b) $\hat{\beta} = 0.999$. Robust SE 0.0284, clustered SE 0.0298.

(c) D is randomised within cluster, so $\hat{\beta} \approx \frac{1}{C}\sum_c \eta_c$ plus within-cluster noise, and

$$\mathrm{Var}(\hat{\beta}) \approx \frac{\mathrm{Var}(\eta_c)}{C} + \text{within-cluster sampling noise.}$$

The robust SE only captures the second term; the clustered SE captures both. With $\sigma_\eta^2 = 1$ the first term is $\approx 1/\sqrt{100} = 0.10$, essentially the whole clustered SE. With $\sigma_\eta^2 = 0$ it vanishes and the two coincide. The cluster intercepts ν_c play no role: they are orthogonal to the randomised D_{ic}. It is heterogeneity in the treatment effect that makes clustering matter.

(d) Rejection rates of $H_0 : \beta = 1$ at 5%: robust 54.0%, clustered 4.7%, block bootstrap 5.0%.

(e) With $\{\eta_c\}$ fixed, testing $H_0 : \beta = \frac{1}{100}\sum_c \eta_c$: robust 4.6%, clustered 0.0%, block bootstrap 0.0%.

(f) In (d) the target is the population average treatment effect across sites, and the 100 sites are a sample: the clustered SE is correctly sized, the robust one is not. In (e) the target is the sample average treatment effect for these 100 sites: the η_c are fixed, the robust SE is correctly sized and the clustered one is very conservative. The appropriate level of clustering depends on the estimand. In an experiment randomised at the individual level, one can report both: the gap between them reflects treatment effect heterogeneity across sites.

Question 5: Bootstrap OLS

Testing $\beta_D = 0$ with 20 clusters

	Method	p-value
OLS (Question 5)		
(b)	Clustered SE, t_{19}	0.0076
(c)	Block bootstrap SE	0.0119
(d)	Block bootstrap percentile-t	0.0280
(e)	Wild cluster bootstrap, null imposed	0.0120
Probit (Question 6)		
(a)	Clustered SE, t_{19}	0.0124
(b)	Block bootstrap percentile-t	0.0211
(c)	Cluster-robust score test	0.0173
(d)	Score wild bootstrap	0.0130

(a) $\hat{\beta}_D = 0.2762$, clustered SE 0.0925.

(b) $t = 2.987$; referring it to t_{19}, $p = 0.0076$. We reject $\beta_D = 0$.

(c) Resampling the 20 clusters 1,000 times, the standard deviation of $\hat{\beta}_D^*$ is 0.0993, giving $p = 0.0119$.

(d) In each bootstrap sample, recompute the clustered SE treating each drawn cluster as a new cluster (`idcluster()`), and form $t^* = (\hat{\beta}_D^* - \hat{\beta}_D)/\mathrm{se}_{cl}^*$, centred at the sample estimate. The symmetric percentile-t p-value is $\mathbb{P}^*\big(|t^*| \geq |t|\big) = 0.0280$.

(e) Wild cluster bootstrap with the null imposed:

1. Estimate the restricted model $y = a + cX + dX2 + \tilde{u}$ (no D).
2. For each replication draw one Rademacher weight $w_g \in \{-1, +1\}$ per cluster and set $y_i^* = \hat{y}_i^R + w_{g(i)}\tilde{u}_i$.
3. Re-estimate the unrestricted model on y^* and compute $t^* = \hat{\beta}_D^*/\mathrm{se}_{cl}^*$.
4. $p = \mathbb{P}^*(|t^*| \geq |t|) = 0.0120$.

(f) All reject at 5%. With only 20 clusters, (b) relies on large-G asymptotics and tends to over-reject. (c) replaces the standard error but keeps the asymptotic reference distribution, so it gives no refinement. (d) is asymptotically refined, but with a binary outcome, a binary regressor and 20 clusters the resampled datasets can be near-degenerate, making se_{cl}^* unstable. The wild bootstrap in (e) does not resample observations, imposes the null, and performs well with few clusters (Cameron, Gelbach and Miller, 2008): it is the most reliable here.

Question 6: Bootstrap Probit

(a) $\hat{\beta}_D = 1.1346$, clustered SE 0.4107, $t = 2.762$, $p = 0.0124$ (against t_{19}).

(b) Block bootstrap of the clustered t-statistic as in Question 5(d), re-estimating the probit in each replication: $p = 0.0211$. Six of the 1,000 replications fail and are dropped.

(c) Write $\theta = (\beta_D, \gamma)$, with γ the constant and the coefficients on X and $X2$. Estimate the probit under the null (y on X, $X2$), giving $\tilde{\theta} = (0, \tilde{\gamma})$. The score of observation i is

$$s_i(\theta) = u_i(\theta)\, x_i, \qquad u_i(\theta) = \frac{\phi(x_i'\theta)\big(y_i - \Phi(x_i'\theta)\big)}{\Phi(x_i'\theta)\big(1 - \Phi(x_i'\theta)\big)}.$$

The restricted first-order conditions set $\sum_i s_{\gamma,i}(\tilde{\theta}) = 0$, so the test is about $\sum_i s_{D,i}(\tilde{\theta})$ (i.e. the D-component of the score). A first-order expansion around $\theta_0 = (\beta = 0, \gamma_0)$ gives

$$\sum_i s_{D,i}(\tilde{\theta}) \approx \sum_i r_i(\theta_0), \qquad r_i = s_{D,i} - H_{D\gamma} H_{\gamma\gamma}^{-1}\, s_{\gamma,i},$$

with H the Hessian of the log-likelihood. Under the null each r_i has mean zero and clusters are independent, so with $r_g = \sum_{i \in g} \hat{r}_i$ (sum residuals within clusters),

$$t_S = \frac{\sum_g r_g}{\sqrt{\sum_g r_g^2}} \xrightarrow{d} N(0,1), \qquad LM = t_S^2 \xrightarrow{d} \chi_1^2.$$

$LM = 5.66$, $p = 0.0173$.

(d) Score wild bootstrap (Kline and Santos, 2012). Keep $\tilde{\theta}$, the r_g and H from (c) fixed. For $b = 1, \ldots, 1000$, draw one Rademacher weight $w_g \in \{-1, +1\}$ per cluster and compute

$$t_S^{*b} = \frac{\sum_g w_g r_g}{\sqrt{\sum_g (w_g r_g)^2}} = \frac{\sum_g w_g r_g}{\sqrt{\sum_g r_g^2}}.$$

The symmetric percentile-t p-value is $\mathbb{P}^*\big(|t_S^*| \geq |t_S|\big) = 0.0130$. The model is never re-estimated, so no replication fails.

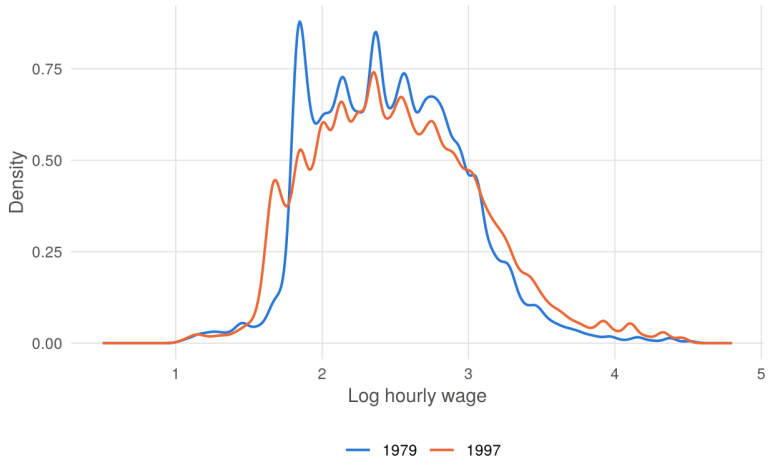
(e) Similarly to Q5, a) requries asymptotics to work. With 20 clusters it's too optimisitic. b) works in principle but with 20 clusters and binary outcomes some repetitions can fail. c) and d) use the same statistic. c), however, requires convergence to the asymptotic distribution of the tstat. In d), we use the bootstrapped tstat distribution instead. Lastly, d) does not require re-estimation so it doesn't fail and it's computationally cheap. It is probably the way to go here given the small number of clusters.

Question 7: Reweighting

(a) The main decisions in the cleaning programs:

- Keep workers with 0–39 years of potential experience.
- Top-coded earnings are multiplied by 1.5.
- Hourly wages below \$1.675 (1982 dollars) are dropped, as are wages above 1/35th of the top-coded weekly earnings.
- Observations are weighted by hours worked times the sampling weight (`wgt_hrs`), so the densities describe the distribution of wages per hour worked.
- Most analyses are restricted to full-time, full-year workers.

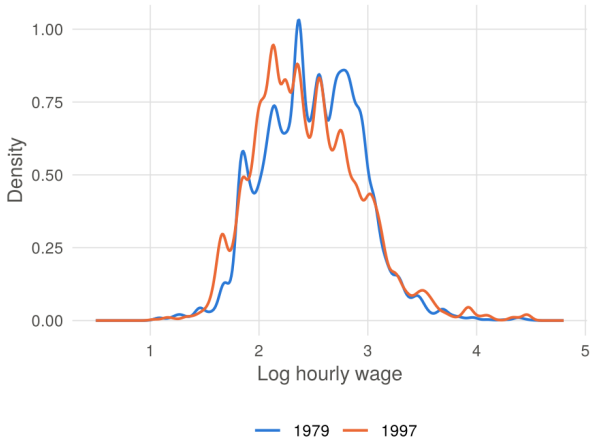
(b) 131,089 observations in 1979 and 109,273 in 1997 (`hr_wage_sample==1`, positive weight).



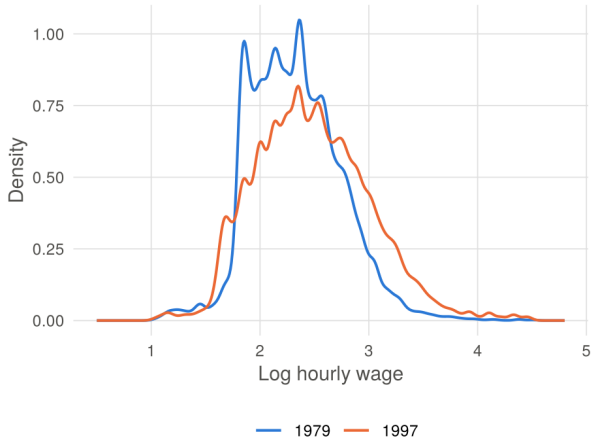
Log hourly wages, 1979 and 1997

(c) The 1997 distribution is more dispersed: the standard deviation rises from 0.50 to 0.57 and the 90–10 gap from 1.22 to 1.44. Most of the widening is in the upper half (90–50 gap +0.13, 50–10 gap +0.08). The spike at the bottom, the minimum wage, moves left as its real value falls.

(d)



Black men, 25–50



White women, 25–50

For black men the distribution shifts left (mean log wage 2.50 to 2.45), with mass moving out of the upper middle. For white women it shifts right (2.33 to 2.48) and widens. The two groups move in opposite directions.

(e) Let X be the demographics and $p(X) = \mathbb{P}(t = 97 \mid X)$, $p = \mathbb{P}(t = 97)$. The 1997 wage distribution with the 1979 composition is

$$f^C(w) = \int f(w \mid X, t = 97)\, dF(X \mid t = 79) = \int f(w \mid X, t = 97)\, \psi(X)\, dF(X \mid t = 97),$$

where, by Bayes' rule,

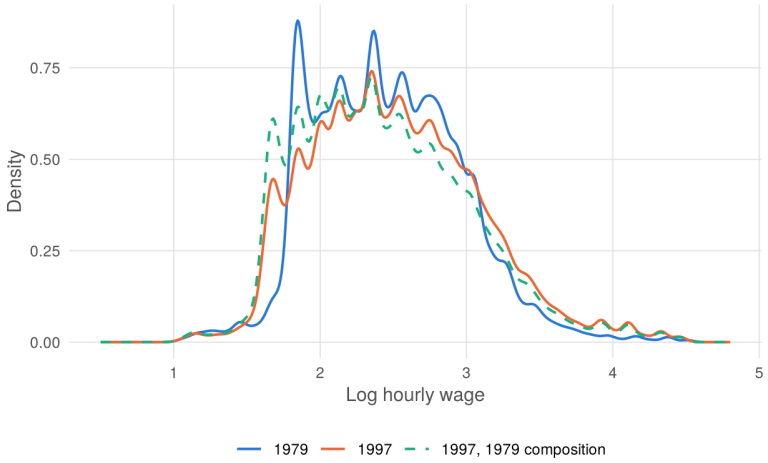
$$\psi(X) = \frac{dF(X \mid t = 79)}{dF(X \mid t = 97)} = \frac{1 - p(X)}{p(X)} \cdot \frac{p}{1 - p}.$$

$p(X)$ comes from the logit below, weighted by `wgt_hrs`.

Logit of the 1997 dummy

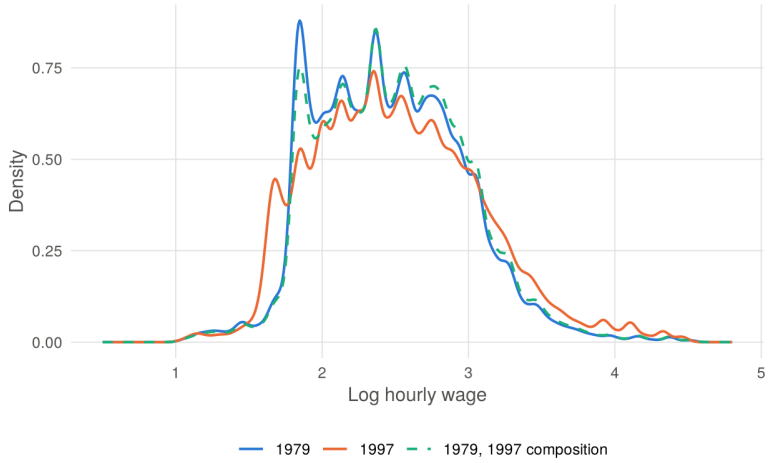
	Coefficient
Female	0.1958
	(0.0096)
Black	0.0394
	(0.0170)
Other race	0.6265
	(0.0276)
Age	0.1655
	(0.0026)
Age^2	-0.0020
	(0.0000)
Constant	-3.0528
	(0.0471)
Observations	240,362

(f) Weight the 1997 observations by `wgt_hrs` $\times \hat{\psi}(X)$.



1997 reweighted to the 1979 demographic composition

(g) Weight the 1979 observations by `wgt_hrs` $\times \hat{\psi}(X)^{-1}$, with $\psi^{-1}(X) = \frac{p(X)}{1-p(X)} \cdot \frac{1-p}{p}$.



1979 reweighted to the 1997 demographic composition

(h) Demographic change explains little of the rise in dispersion. With the 1979 composition, the 1997 90–10 gap would be 1.45 (actual: 1.44); with the 1997 composition, the 1979 gap would be 1.26 (actual: 1.22). At most about a sixth of the 0.21 increase is compositional. Composition does account for the rise in the mean: at the 1979 composition the 1997 mean (2.44) is below the actual 1979 mean (2.46). The reweighting only uses sex, race and age, not education.

(i) –(j)

Age moments: 1979, 1997, and 1997 reweighted to 1979

	1979	1997	1997 reweighted
All workers			
Mean	35.33	37.45	35.17
SD	12.55	11.22	12.62
p10	21	23	20
p25	25	28	24
p50	33	37	33
p75	45	46	45
p90	55	53	54
Blacks			
Mean	35.32	36.62	34.34
SD	11.97	10.81	12.00
p10	21	23	20
p25	26	28	24
p50	33	36	32
p75	44	45	43
p90	54	52	52

For all workers the reweighted 1997 age distribution matches 1979 closely, within a year at every percentile. For blacks it does not: the reweighted mean is a year too low and the 25th and 90th percentiles are two years off. The logit only shifts the intercept for blacks, so it cannot match their age distribution.

(k)

Age moments of 1997 reweighted to 1979, by age specification

	1979	Linear	Quadratic	Quartic	Age FE	Age FE × race
All workers						
Mean	35.33	35.57	35.17	35.29	35.30	35.30
SD	12.55	10.97	12.62	12.56	12.55	12.55
p10	21	22	20	20	20	20
p25	25	27	24	25	25	25
p50	33	35	33	33	33	33
p75	45	43	45	45	45	45
p90	55	51	54	55	55	55
Blacks						
Mean	35.32	34.88	34.34	34.36	34.37	35.31
SD	11.97	10.50	12.00	11.93	11.92	11.95
p10	21	22	20	21	21	21
p25	26	26	24	25	25	26
p50	33	34	32	32	32	33
p75	44	42	43	43	43	44
p90	54	49	52	53	53	54

(l) A linear term in age is clearly inadequate. For all workers the quadratic is close and a quartic or full set of age dummies matches 1979 almost exactly. For blacks no specification in age alone helps; balance requires interacting age with race. The propensity score has to be flexible in the subgroups one wants to compare, and balance should be checked within them.

Question 8: Duflo, Hanna and Ryan (2012)

Periods are indexed $t \in \{1, 2\}$, as in the rest of the question.

(a) The state in period 2 is L_1 (whether the bonus is still available) and the shock ε_2. Consumption is $C_2 = 5 + 5 \cdot \mathbf{1}\{L_1 = 0,\, L_2 = 0\}$, so

$$V_2(L_1, \varepsilon_2) = 5\beta + \max\big\{\, 5\beta \cdot \mathbf{1}\{L_1 = 0\}\,,\ \mu + \varepsilon_2 \,\big\}.$$

(b) For $\varepsilon \sim N(0,1)$ and a constant a, using $\mathbb{E}[\varepsilon\, \mathbf{1}\{\varepsilon > c\}] = \phi(c)$,

$$m(a) \equiv \mathbb{E}\big[\max\{a,\ \mu + \varepsilon\}\big] = a\,\Phi(a - \mu) + \mu\,\Phi(\mu - a) + \phi(\mu - a).$$

So $EV_2(1) = 5\beta + \mu\Phi(\mu) + \phi(\mu)$ and $EV_2(0) = 5\beta + m(5\beta)$, and

$$\Delta(\beta,\mu) \equiv EV_2(0) - EV_2(1) = 5\beta\,\Phi(5\beta-\mu) + \mu\,\Phi(\mu-5\beta) + \phi(\mu-5\beta) - \mu\Phi(\mu) - \phi(\mu) \;\geq 0$$

is the option value of keeping the bonus alive. Since $C_1 = 0$,

$$V_1(\varepsilon_1) = \max\big\{(\mu+\varepsilon_1) + \delta\,EV_2(1),\quad \delta\,EV_2(0)\big\},$$

and the teacher takes leisure in period 1 iff $\mu + \varepsilon_1 > \delta\,\Delta(\beta,\mu)$.

(c) With independent shocks,

$$p_{11} = \Phi(\mu-\delta\Delta)\,\Phi(\mu), \qquad p_{10} = \Phi(\mu-\delta\Delta)\,\Phi(-\mu),$$
$$p_{01} = \Phi(\delta\Delta-\mu)\,\Phi(\mu-5\beta), \qquad p_{00} = \Phi(\delta\Delta-\mu)\,\Phi(5\beta-\mu),$$

where $p_{l_1 l_2} = \mathbb{P}(L_1 = l_1, L_2 = l_2)$. The sample log-likelihood is $\ell(\beta,\mu,\delta) = \sum_i \log p_{L_{1i}L_{2i}}$.

(d) The data are 800 draws from a four-cell multinomial (3 free probabilities), so the model with three parameters is exactly identified:

$$\hat\mu = \Phi^{-1}\big(\hat{\mathbb{P}}(L_2{=}1 \mid L_1{=}1)\big), \qquad \hat\beta = \frac{\hat\mu - \Phi^{-1}\big(\hat{\mathbb{P}}(L_2{=}1 \mid L_1{=}0)\big)}{5}, \qquad \hat\delta = \frac{\hat\mu - \Phi^{-1}\big(\hat{\mathbb{P}}(L_1{=}1)\big)}{\Delta(\hat\beta,\hat\mu)},$$

Column (1) of the table.

Duflo, Hanna and Ryan, $N = 800$

	(1) Full model	(2) Myopic
β	0.1678	0.1485
	(0.0186)	(0.0161)
μ	0.3828	0.2864
	(0.0592)	(0.0357)
δ	0.3524	0
	(0.1637)	
Log-likelihood	-1053.639	-1055.743

(e) Column (2) imposes $\delta = 0$. $LR = 2(-1053.639 + 1055.743) = 4.21$ on 1 degree of freedom, $p = 0.040$: we reject myopia at 5%. 0 is at the boundary of the parameter space, so you can apply the appropriate corrections when testing.

δ is identified by comparing period-1 leisure with period-2 leisure among teachers who already lost the bonus. If $\delta = 0$, the teacher ignores the bonus she gives up by shirking in period 1, so she takes leisure iff $\mu + \varepsilon_1 > 0$. A teacher who took leisure in period 1 has no bonus left, so in period 2 she

also takes leisure iff $\mu + \varepsilon_2 > 0$; since the shocks are assumed iid, conditioning on $L_1 = 1$ is not informative on ε_2. Myopia therefore implies $\mathbb{P}(L_1 = 1) = \mathbb{P}(L_2 = 1 \mid L_1 = 1) = \Phi(\mu)$. In the data period-1 leisure is lower, $\hat{\mathbb{P}}(L_1 = 1) = 0.591$ against $\hat{\mathbb{P}}(L_2 = 1 \mid L_1 = 1) = 0.649$, and only the option value $\delta\Delta$ of keeping the bonus can explain that difference.

(f) With $\mu_i \sim N(\mu_1, \sigma_1^2)$, constant within teacher,

$$\mathbb{P}(L_1 = l_1, L_2 = l_2) = \int \Phi\big(\varsigma_1(\mu - \delta\Delta(\beta, \mu))\big)\, \Phi\big(\varsigma_2(\mu - 5\beta(1 - l_1))\big)\, dF_{\mu_1,\sigma_1}(\mu), \qquad \varsigma_t = 2l_t - 1,$$

simulated with S fixed draws $\mu_s = \mu_1 + \sigma_1\tau_s$, recomputing Δ at each draw. The most important difference in this version of the model is that the choice of working in period 1 can be attributed to either high δ or just a persistent high preference for work. With only three probabilities to target (four moments that add to one, so it's three) and four parameters, these are observationally equivalent: the model is not identified, and one parameter has to be fixed. δ is the obvious one, since it has a natural economic interpretation. Fixing δ at different values and maximising over the other parameters:

Random μ: estimates at fixed δ

δ	β	μ_1	σ_1	Log-likelihood
0.05	0.1117 (0.0289)	0.2599 (0.0460)	0.4930 (0.1612)	-1053.639
0.10	0.1182 (0.0307)	0.2733 (0.0476)	0.4579 (0.1748)	-1053.639
0.15	0.1255 (0.0327)	0.2886 (0.0497)	0.4170 (0.1933)	-1053.639
0.20	0.1337 (0.0350)	0.3060 (0.0528)	0.3680 (0.2202)	-1053.639
0.25	0.1432 (0.0377)	0.3261 (0.0571)	0.3067 (0.2650)	-1053.639
0.30	0.1542 (0.0409)	0.3499 (0.0638)	0.2230 (0.3640)	-1053.639
0.35	0.1671 (0.0447)	0.3801 (0.1021)	0.0481 (1.6682)	-1053.639
0.50	0.1708 (0.0183)	0.4200 (0.0426)	0.0000	-1054.045
0.90	0.1647 (0.0184)	0.4924 (0.0491)	0.0000	-1058.706

For $\delta > 0.35$ $\hat{\sigma}_1$ hits zero (boundary solution so no valid SEs), the model collapses to the one without heterogeneity, and the fit gets worse. For values below that, you can see the value of the likelihood doesn't change: indeed, higher δ and lower σ_1 are observationally equivalent. You need to make assumptions about one parameter to make inference (for example calibrating the discount factor from a different setting/data). Standard errors are from the inverse of the numerical Hessian of the simulated log-likelihood at the optimum (like you would in regular MLE). We can do this here because the model is correctly specified by assumption.